Statistical Learning and Machine Learning

Exam Topics - January 2023

TOPIC 1

You are given $N = 6$ data points, $\{x_i\}_{i=1}^{6}$, the values of which are shown in the Figure 1. The data points reside in the 1D space i.e., $x_i \in \mathbb{R}$. It is further known that these points are independent and identically distributed according to the Gaussian distribution. You are asked to estimate the parameters of that Gaussian distribution from the above data points by using the maximum likelihood (ML) approach and the Bayesian approach. Please answer the following questions.

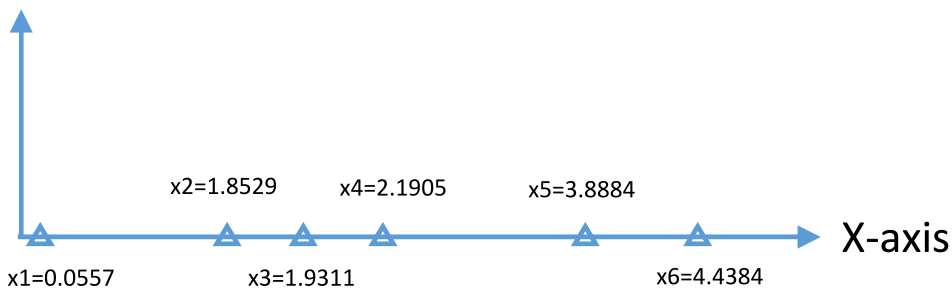

Fig. 1. Data points $\{x_i\}_{i=1}^{6}$ residing in the 1D space obtained from a Gaussian distribution with unknown mean μ and variance σ^2.

1) Estimate the mean of the Gaussian distribution, μ_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 15%)
2) Estimate the variance of the Gaussian distribution. σ^2_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 15%)
3) Comment on whether the above estimates of the mean μ_{ML} and the variance σ^2_{ML} are biased or unbiased? (Score: 10%)
4) If one or both the above estimates obtained in the previous step are biased, how would you make those unbiased? (Score: 10%)
5) Now, assume that the actual variance of the Gaussian distribution that led to the data points in Fig. 1 is known and is $\sigma^2 = 1$. Using that information and the same data points as in Fig. 1, estimate the mean of the Gaussian distribution μ_{Bayes} through the Bayesian approach by using the following conjugate form for the prior distribution of mean: $p(\mu) = \mathcal{N}(\mu|\mu_0 = 4, \sigma_0^2 = 3)$. (Score: 25%)
6) Given that the information about the prior density $p(\mu)$ is reasonable, which of the above density estimation approaches (i.e., ML or Bayesian) do you expect will perform well in the following settings: a) for smaller values of N (e.g., for $N = 6$ as in Fig. 1); b) for $N \to \infty$. (Score: 10%)
7) Briefly mention the reasons behind your answers in the previous question (6). (Score: 15%)

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TOPIC 2

You have a high-dimensional data set on which you want to run simple linear regression (without any regularization) with a mean squared error loss. There is no regression line that fits the data perfectly. To obtain better predictions, you decide to run principal component analysis (PCA) before your regression.

You notice that some of your input features are measured on very different scales. Furthermore, some input features are highly correlated. Please answer the following questions.

1) (Score: 33%) Should you standardize your inputs to be mean zero and have unit variance before running PCA? Yes/no? Why/why not?
2) (Score: 33%) Do you need to remove some of the highly correlated features before you run PCA? Yes/no? Why/why not?
3) (Score: 33%) Together with your friend, you have computed the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_q > 0$ of the empirical covariance matrix. Now, your friend proposes to let $\lambda = \frac{1}{q}\sum_{i=1}^{q}\lambda_i$ and consider the standard deviation

$$S = \sqrt{\frac{1}{q}\sum_{j=1}^{q}(\lambda_j - \lambda)^2} \tag{1}$$

of the eigenvalues. Suppose $S = 0$. You friend says: "Well, $S = 0$, so we can just use one principal direction". Is this appropriate? Yes/no? Why/why not?

TOPIC 3

Two classes of a binary classification problem are given by the blue triangles and red circles in Figure 2.

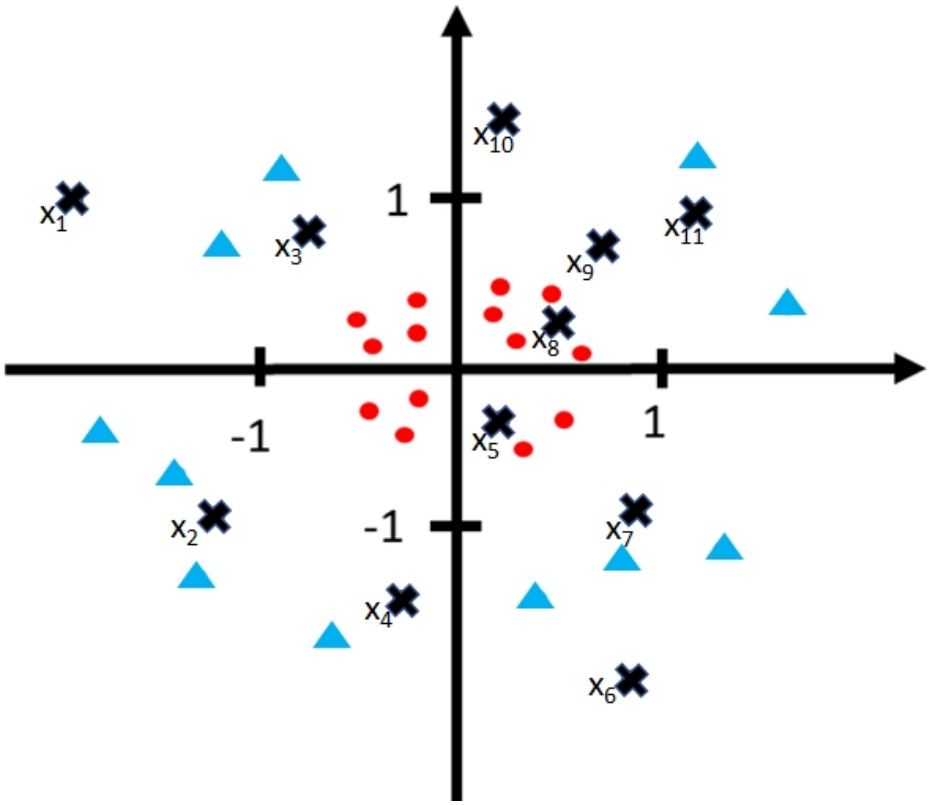

Fig. 2. 2-dimensional data points plotted as dots. Data belonging to class 1 are illustrated as blue triangles. Data belonging to class 2 are illustrated as red circles. Test data $\mathbf{x}_i$, $i = 1, \ldots, 11$ are illustrated as black x's.

Please answer the following questions.

1) (Score: 10%) Can you use a Perceptron to classify data from the two classes? If yes, how this can be done? If no, why not?
2) (Score: 10%) Can you use Logistic Regression to classify data from the two classes? If yes, how this can be done? If no, why not?
3) (Score: 10%) Can you use a Support Vector Machine to classify data from the two classes? If yes, how this can be done? If no, why not?
4) (Score: 10%) Can you use Linear Regression (to class indicator matrix) to classify data from the two classes? If yes, how this can be done? If no, why not?
5) (Score: 20%) There exists a data transformation $y_i = \phi(\mathbf{x}_i)$ such that the linear Support Vector Machine applied on the transformed data can provide 100% training classification performance. Write down the data transformation function $y_i = \phi(\mathbf{x}_i)$.
6) (Score: 15%) Draw the transformed data (resulted using the transformation in question 5), the decision function of a linear Support Vector Machine applied on the transformed data in the new feature space, and the corresponding decision function in the original feature space. When drawing the transformed data, no exact values of their dimension(s) are needed, just the rough arrangement of them in the new feature space.
7) (Score: 15%) Show which are the Support Vectors (a) in the new feature space (drawn to answer in question 6), (b) in the original feature space.
8) (Score: 10%) To which class will the points $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{11}$ be classified using the SVM classifier you used to answer questions 5-7?

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TOPIC 4

You are given a set of data points $\mathbf{x}_i \in \mathbb{R}^8$, $i = 1, \ldots, N$, each followed by a four-dimensional target $\mathbf{t}_i \in \mathbb{R}^4$, $i = 1, \ldots, N$. You want to use a feedforward neural network formed by three hidden-layers for solving this regression problem.

Please answer the following questions (give short, 2-4 line responses).

1) (Score: 15%) Describe which are the parameters and the hyper-parameters of the neural network. Use appropriate mathematical symbols (e.g., matrices, functions).
2) (Score: 15%) Draw a neural network that can be used to solve the above-described classification problem. Make appropriate decisions for <u>all</u> the hyper-parameters of the neural network. In your drawing include also <u>all</u> the parameters of the neural network and the activation functions.
3) (Score: 15%) Describe how to estimate the values of the neural network parameters and its hyper-parameters.
4) (Score: 15%) Write down which optimization problem (loss function) you will use to train the network, and describe all terms (including the dimensions of each matrix/vector). Why is this optimization problem used when we optimize the neural network for solving a regression problem?
5) (Score: 10%) Show why using the linear activation function (for all layers of the network) makes the above neural network not capable to achieve good performance in nonlinear regression problems. To do this, write down the equivalent, but simplified, neural network for this case, and draw it. Assume that the bias values for all neurons are equal to zero.
6) (Score: 15%) If you were going to use linear activation functions for all the layers of the neural network, how would you optimize the parameters of the neural network? Explain why you would have to use that optimization process.
7) (Score: 15%) How would the neural network trained following the process in question 6 compare to using a linear regression model?